

DETROIT ST CLAIR SHORES, MI, USA

WMO: 997996

Lat: 42.4711N

Lon: 82.8766W

Elev: 591

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.0	(c) 7.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 26.3	(k) 23.9	(l) 24.2	(m) 22.8	(n) 8.5	(o) 230	(p) 0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.0	(c) 88.7	(d) N/A	(e) 85.5	(f) N/A	(g) 82.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 11.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.4	(b) 20.7	(c) 18.5	(e) DB	(f) -3.7	(g) 94.1	(h) 8.3	(i) 2.9	(j) -9.7	(k) 96.2	(l) -14.5	(m) 97.9	(n) -19.2	(o) 99.6	(p) -25.2	(q) 101.7	(r) N/A
(4) 24.4	(5) 20.7	(6) 18.5	(7) DB	(8) -3.7	(9) 94.1	(10) 8.3	(11) 2.9	(12) -9.7	(13) 96.2	(14) -14.5	(15) 97.9	(16) -19.2	(17) 99.6	(18) -25.2	(19) 101.7	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	24.2	25.5	34.3	47.5	59.6	68.7	74.0	71.9	65.6	53.6	41.2	32.2		
	DBStd	19.22	10.72	10.09	9.53	7.75	7.93	6.53	5.87	4.86	6.75	7.12	8.61	8.14		
	HDD50	3016	802	687	496	139	10	0	0	0	0	43	285	554		
	HDD65	6270	1266	1107	953	527	209	34	4	5	74	358	715	1018		
	CDD50	3010	1	1	8	63	307	561	745	680	467	156	20	1		
	CDD65	788	0	0	0	1	41	144	285	221	91	5	0	0		
	CDH74	4668	0	0	0	29	260	830	1992	1148	394	15	0	0		
	CDH80	1224	0	0	0	3	61	255	582	224	97	2	0	0		
	WSAvg	9.0	9.7	9.3	9.8	10.5	8.9	8.5	7.8	7.8	8.2	9.4	9.3	9.4		
	PrecAvg	35.0	2.3	2.1	2.5	3.3	3.6	3.3	3.5	3.1	3.4	2.6	2.6	2.7		
(15) Precipitation	PrecMax	53.2	4.4	5.3	5.6	5.7	7.0	6.1	7.8	5.8	7.2	6.1	6.5	4.8		
	PrecMin	27.0	0.7	0.5	0.8	0.9	1.4	1.5	1.1	0.6	0.9	0.5	0.6	0.9		
	PrecStd	5.4	1.0	1.1	1.2	1.3	1.6	1.3	1.7	1.5	1.9	1.2	1.2	0.9		
	PrecStd	5.4	1.0	1.1	1.2	1.3	1.6	1.3	1.7	1.5	1.9	1.2	1.2	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	56.2	64.4	79.1	86.1	91.7	94.8	89.2	89.6	75.8	65.5	57.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	50.6	49.1	58.9	70.6	80.7	86.5	89.2	85.5	82.0	70.1	60.3	52.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	43.3	43.1	53.3	65.3	76.4	82.6	86.0	82.4	78.0	67.5	57.0	47.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	38.0	37.6	47.6	59.5	71.8	78.4	83.0	79.9	75.4	64.2	53.4	42.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.1	12.9	14.0	14.2	12.7	13.0	12.1	12.5	12.1	10.7	9.3		
		MCDBR	16.9	18.7	20.4	23.3	19.8	17.5	16.5	14.9	15.3	15.2	16.0	15.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.334	0.352	0.397	0.428	0.445	0.435	0.426	0.388	0.360	0.321	0.308		
	taud		2.388	2.288	2.352	2.246	2.213	2.202	2.255	2.269	2.356	2.418	2.490	2.425		
	Ebn at Noon		265	272	283	276	269	264	266	264	267	261	258	252		
	Edh at Noon		25	33	35	42	44	45	43	41	35	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		457	738	1113	1470	1720	1946	1937	1679	1354	880	542	386		
	RadStd		51	81	112	129	147	120	96	68	117	75	69	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page